Agar

1 Nonproprietary Names

JP: Agar
PhEur: Agar
USP-NF: Agar

2 Synonyms

Agar-agar; agar-agar flake; agar-agar gum; Bengal gelatin; Bengal gum; Bengal isinglass; Ceylon isinglass; Chinese isinglass; E406; gelosa; gelose; Japan agar; Japan isinglass; layor carang.

3 Chemical Name and CAS Registry Number

Agar [9002-18-0]

4 Empirical Formula and Molecular Weight

See Section 5.

5 Structural Formula

Agar is a dried, hydrophilic, colloidal polysaccharide complex extracted from the agarocytes of algae of the Rhodophyceae. The structure is believed to be a complex range of polysaccharide chains having alternating α-(1→3) and β-(1→4) linkages. There are three extremes of structure noted: namely neutral agarose; pyruvated agarose having little sulfation; and a sulfated galactan. Agar can be separated into a natural gelling fraction, agarose, and a sulfated nongelling fraction, agaropectin.

6 Functional Category

Emulsifying agent; stabilizing agent; suppository base; suspending agent; sustained-release agent; tablet binder; thickening agent; viscosity-increasing agent.

7 Applications in Pharmaceutical Formulation or Technology

Agar is widely used in food applications as a stabilizing agent. In pharmaceutical applications, agar is used in a handful of oral tablet and topical formulations. It has also been investigated in a number of experimental pharmaceutical applications including as a sustained-release agent in gels, beads, microspheres, and tablets.[1-4] It has also been reported to work as a disintegrant in tablets.[5] Agar has been used in a floating controlled-release tablet; the buoyancy in part being attributed to air entrapped in the agar gel network.[6] It can be used as a viscosity-increasing agent in aqueous systems. Agar can also be used as a base for nonmelting, and nondisintegrating suppositories.[7] Agar has an application as a suspending agent in pharmaceutical suspensions.[8]

8 Description

Agar occurs as transparent, odorless, tasteless strips or as a coarse or fine powder. It may be weak yellowish-orange, yellowish-gray to pale-yellow colored, or colorless. Agar is tough when damp, brittle when dry.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for agar.

Test	JP XV	PhEur 6.3	USP32–NF27
Identification	+	+	+
Characters	+	+	—
Swelling index	—	+	—
Arsenic	—	—	⩽3 ppm
Lead	—	—	⩽0.001%
Sulfuric acid	+	—	—
Sulfurous acid and starch	+	—	—
Gelatin	—	+	+
Heavy metals	—	—	⩽0.004%
Insoluble matter	⩽15.0 mg	⩽1.0%	⩽15.0 mg
Water absorption	⩽75 mL	—	⩽75 mL
Loss on drying	⩽22.0%	⩽20.0%	⩽20.0%
Microbial contamination	—	⩽10^3 cfu/g[a]	+
Total ash	⩽4.5%	⩽5.0%	⩽6.5%
Acid-insoluble ash	⩽0.5%	—	⩽0.5%
Foreign organic matter	—	—	⩽1.0%
Limit of foreign starch	—	—	+

(a) Total viable aerobic count, determined by plate-count.

10 Typical Properties

NIR spectra *see* Figure 1.

Solubility Soluble in boiling water to form a viscous solution; practically insoluble in ethanol (95%), and cold water. A 1% w/v aqueous solution forms a stiff jelly on cooling.

11 Stability and Storage Conditions

Agar solutions are most stable at pH 4–10.

Agar should be stored in a cool, dry, place. Containers of this material may be hazardous when empty since they retain product residues (dust, solids).

12 Incompatibilities

Agar is incompatible with strong oxidizing agents. Agar is dehydrated and precipitated from solution by ethanol (95%). Tannic acid causes precipitation; electrolytes cause partial dehydration and decrease in viscosity of sols.[9]

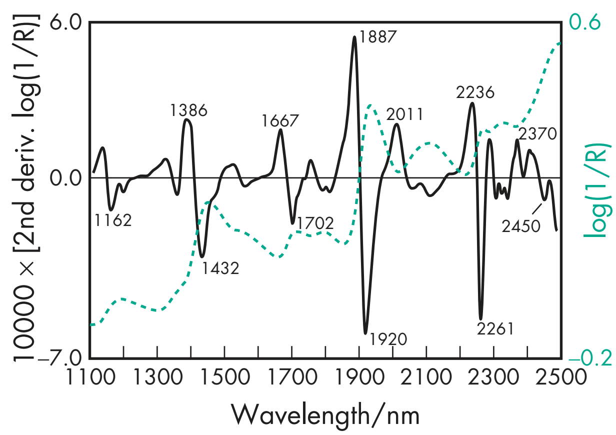

Figure 1: Near-infrared spectrum of agar measured by reflectance.

13 Method of Manufacture

Agar is obtained by freeze-drying a mucilage derived from *Gelidium amansii* Lamouroux, other species of the same family (Gelidiaceae), or other red algae (Rhodophyta).

14 Safety

Agar is widely used in food applications and has been used in oral and topical pharmaceutical applications. It is generally regarded as relatively nontoxic and nonirritant when used as an excipient.

LD_{50} (hamster, oral): 6.1 g/kg[10]

LD_{50} (mouse, oral): 16.0 g/kg

LD_{50} (rabbit, oral): 5.8 g/kg

LD_{50} (rat, oral): 11.0 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled. When heated to decomposition, agar emits acrid smoke and fumes.

16 Regulatory Status

GRAS listed. Accepted for use as a food additive in Europe. Included in the FDA Inactive Ingredients Database (oral tablets). Included in the Canadian List of Acceptable Non-medicinal Ingredients. Included in nonparenteral medicines licensed in the UK.

17 Related Substances

—

18 Comments

The drug release mechanism of agar spherules of felodipine has been studied and found to follow Higuchi kinetics.[11] Agar has also been used to test the bioadhesion potential of various polymers.[12]

The EINECS number for agar is 232-658-1.

19 Specific References

1 Bhardwaj TJ *et al*. Natural gums and modified natural gums as sustained release carriers. *Drug Dev Ind Pharm* 2000; **26**(10): 1025–1038.
2 Sakr FM *et al*. Design and evaluation of a dry solidification technique for preparing pharmaceutical beads. *STP Pharma Sci* 1995; **5**(4): 291–295.
3 Boraie NA, Naggar VF. Sustained release of theophylline and aminophylline from agar tablets. *Acta Pharm Jugosl* 1984; **34**(Oct–Dec): 247–256.
4 Nakano M *et al*. Sustained release of sulfamethizole from agar beads. *J Pharm Pharmacol* 1979; **31**: 869–872.
5 Fassihi AR. Characteristics of hydrogel as disintegrant in solid dose technology. *J Pharm Pharmacol* 1989; **41**(12): 853–855.
6 Desai S, Boston S. A floating controlled-release drug delivery system: *in vitro–in vivo* evaluation. *Pharm Res* 1993; **10**: 1321–1325.
7 Singh KK *et al*. Studies on suppository bases: design and evaluation of sodium CMC and agar bases. *Indian Drugs* 1994; **31**(April): 149–154.
8 Kahela P *et al*. Effect of suspending agents on the bioavailability of erythromycin ethylsuccinate mixtures. *Drug Dev Ind Pharm* 1978; **4**(3): 261–274.
9 Gennaro AR, ed. *Remington: The Science and Practice of Pharmacy*, 20th edn. Baltimore: Lippincott Williams & Wilkins, 2000; 1030.
10 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 90–91.
11 Patil AK *et al*. Preparation and evaluation of agar spherules of felodipine. *J Pure Appl Microbiol* 2007; **1**(2): 317–322.
12 Bertram U, Bodmeier R. In situ gelling, bioadhesive nasal inserts for extended drug delivery: in vitro characterization of a new nasal dosage form. *Eur J Pharm* 2006; **27**(1): 62–71.

20 General References

—

21 Author

VK Gupta.

22 Date of Revision

10 February 2009.

Albumin

1 Nonproprietary Names

BP: Albumin Solution

PhEur: Human Albumin Solution

USP: Albumin Human

2 Synonyms

Alba; *Albuconn*; *Albuminar*; albumin human solution; albumini humani solutio; *Albumisol*; *Albuspan*; *Albutein*; *Buminate*; human serum albumin; normal human serum albumin; *Octalbin*; *Plasbumin*; plasma albumin; *Pro-Bumin*; *Proserum*; *Zenalb*.

3 Chemical Name and CAS Registry Number

Serum albumin [9048-49-1]

4 Empirical Formula and Molecular Weight

Human serum albumin has a molecular weight of about 66 500 and is a single polypeptide chain consisting of 585 amino acids. Characteristic features are a single tryptophan residue, a relatively low content of methionine (6 residues), and a large number of cysteine (17) and of charged amino acid residues of aspartic acid (36), glutamic acid (61), lysine (59), and arginine (23).

5 Structural Formula

Primary structure Human albumin is a single polypeptide chain of 585 amino acids and contains seven disulfide bridges.

Secondary structure Human albumin is known to have a secondary structure that is about 55% α-helix. The remaining 45% is believed to be divided among turns, disordered, and β structures.[1]

Albumin is the only major plasma protein that does not contain carbohydrate constituents. Assays of crystalline albumin show less than one sugar residue per molecule.

6 Functional Category

Stabilizing agent; therapeutic agent.